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17 The nth term of a sequence is given by $an^2 + bn$ where a and b are integers.

The 2nd term of the sequence is 26
The 3rd term of the sequence is 48

Find the 4th term of the sequence.

$n = 2$: $4a + 2b = 26$

$n = 3$: $9a + 3b = 48$

1 mark
Substitute n = 2 and n = 3.

$\times 3$: $12a + 6b = 78$

$\times 2$: $18a + 6b = 96$

1 mark
Match the b terms to eliminate b.

Subtract: $6a = 18$, so $a = 3$

$12 + 2b = 26$, so $b = 7$

1 mark
Solve for both coefficients.

4th term $= 3 \times 4^2 + 7 \times 4 = 48 + 28 = 76$

1 mark
Use the formula with n = 4.

76

(Total for Question 17 is 4 marks)